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A stage-gate decision process for guiding the development of AI solutions for preventive maintenance

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ABSTRACT

The development of Artificial Intelligence (AI) solutions for preventive maintenance applications is a risky and resource-demanding process. Typically, there are several candidate solutions whose performance in transforming data into useful prognostic information is initially uncertain. These uncertainties can be managed by structuring the development process into multiple stages that help choose and implement the final solution. In this paper, we propose such a stage-gate process by using Robust Portfolio Modelling to screen increasingly detailed candidate solutions through four development stages and three decision gates. The development stages generate evidence on how the candidate solutions contribute to six development objectives that represent different financial and technical criteria. At the decision gates, decisions about the continuation/termination of candidate solutions are taken by identifying portfolios of non-dominated candidate solutions subject to time and budget constraints. Uncertainties are captured by admitting incomplete information about the criteria weights and scores of candidate solutions. We illustrate the process by considering the development of an AI solution for a train's toilet door system. The process brings consistency to the development process and, among other benefits, helps mitigate the risk of missing the development objectives due to premature fixation on a single candidate solution.

1. Introduction

The wide adoption of modern technologies, such as sensors that send data to cloud services, offers exciting opportunities for improving the efficiency of maintaining technical systems based on improved data resources. In Prognostics and Health Management (PHM) systems (Atamuradov et al., 2017; Javed et al., 2017), data can be valuable for estimating the system's Remaining Useful Life (RUL) (Kan et al., 2015) and for the design of optimal maintenance policies (Fink et al., 2020). Accordingly, we use the term *PHM task* to refer to the transformation of data resources into prognostic information that supports maintenance decision-making. In particular, we focus on completing the PHM task with a data-driven *Artificial Intelligence (AI)-based PHM approach* (Tsui et al., 2015).

Yet there are challenges in choosing AI approaches for the development of an operational AI solution (Lu, 2019; Nguyen et al., 2022). First, the performance of the AI solution needs to guarantee the accuracy of the RUL estimates while minimising potential shortcomings, such as false alarms caused by incoherent data, faulty sensors, and system characteristics. Second, it may be impossible to complete the PHM task in time due to computational latency of the AI approach. Third, development costs and durations vary between alternative approaches. Fourth, there may be challenges related to the interpretability of results, generalisation deficiencies, and the data intensity of industrial applications (Biggio and Kastanis, 2020; Theissler et al., 2021). Fifth, the potential economic benefits are not limited to reduced maintenance and operation costs, as there are ancillary benefits, such as building in-house expertise in AI or establishing new collaboration opportunities. Given the breadth of relevant factors, developing an AI solution for PHM is a dynamic multi-criteria problem that can benefit from a structured decision process. From the viewpoint of decision support, it is helpful to formulate and elaborate the *development objectives* that guide this process towards a solution for a specific PHM task, because there is currently no universal AI approach that can be applied to all PHM tasks (Lu, 2019).

Previous research in this area has not proposed a structured decision-making framework that can accommodate both technical and organisational criteria in the early phases of the development of an AI-based solution for the PHM task. Thus, the main contribution of this paper is the elaboration of a structured stage-gate process for guiding such a development. In this process, the series of development

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stages facilitates cumulative learning and a reduction of uncertainties by producing a growing and tested body of evidence on the performance of the *candidate solutions* (hereafter referred to as *candidates*) under development. This stage-gate process can be particularly useful, because the field of AI evolves fast (Lu, 2019). Thus, the information in the early development stages can be quite incomplete in the case of AI, wherefore it may be necessary to consider many candidates in parallel. In this setting, structuring this process as a funnel helps advance several potential candidates towards an AI-based solution, making it possible to mitigate the risk of prematurely focusing on a single candidate and build competence without a strong prior background in AI algorithms or the need to share data resources with outside experts.

Specifically, this iterative stage-gate process starts with the specification of *use cases* that describe how a well-defined *PHM task* is to be completed by leveraging an AI-based prognostic approach on available *data resources*. It proceeds through the stages of exploratory testing and detailed implementation planning to the eventual implementation of the *AI solution*. The stages are separated by decision gates at which the decisions for the future development of candidates are made. At the decision gates, multiple objectives – operationalised in terms of corresponding criteria and measurement scales – are considered by assessing the candidates by drawing on complementary sources of information, including expert judgements. Methodologically, the screening of candidates is facilitated with Robust Portfolio Modelling (RPM) (Liesiö et al., 2007, 2008). The advantage of using RPM is that it handles incomplete information, identifies robust portfolios under constraints, and facilitates transparent decision-making under uncertainty. After the early decision gates, the chosen portfolio is not a final set of AI solutions, but a set of candidates for further development. Because PHM tasks are tightly linked to specific data resources and operational constraints, managing these multiple candidates in parallel helps the organisation learn about AI approaches as well as task-specific PHM data integration challenges.

This paper is structured as follows. Section 2 discusses AI approaches from three viewpoints and motivates the proposed stage-gate process outlined in Section 3. Section 4 presents an illustrative case study. Section 5 discusses the advantages and limitations of the process, as well as future research needs. Section 6 concludes.

2. Literature review on the applicability of AI in PHM

2.1. AI in prognostics

The number of citations to publications on applications of AI techniques in PHM has grown exponentially over the past decade (Carvalho et al., 2019). A large variety of applications have been reported, including high-speed trains (e.g. Atamuradov et al., 2017), rail tracks (e.g. Chenariyan Nakhaee et al., 2019), electrical, electronic, mechanical, and control engineering (e.g. Nguyen et al., 2022), rotating machinery operating under nonlinear and non-stationary conditions (e.g. Kan et al., 2015), wind turbines (e.g. Leite et al., 2018) autonomous ships (e.g. Ellefsen et al., 2019), rolling element bearings (e.g. Hamadache et al., 2019), to cite a few. Furthermore, applications exist at component, subsystem, and system levels (Javed et al., 2017).

AI approaches can handle high-dimensional, multivariate, and possibly unstructured data in complex and dynamically evolving environments (Wuest et al., 2016). However, as in other domains (Uren and Edwards, 2023), key considerations in developing AI solutions for PHM pertain to the availability of knowledge, information, and data. More broadly, Uren's model for AI adoption (Uren and Edwards, 2023) merges technology readiness with four intertwined socio-technical factors: people, data, process, and technology, and underlines the evolving role of data during adoption. Issues of data quality can hinder AI adoption (Chenariyan Nakhaee et al., 2019).

PHM systems generate diverse data types, quantities, and qualities. Data may also be associated with various labels which are not necessarily standardised or may be of low quality due to a lack of commitment by maintenance operators (Abid et al., 2021). Such characteristics impact the compatibility between AI approaches and data resources. For example, Nguyen et al. (2022) outline how prior knowledge, expert knowledge, data volume, and data properties affect the choice of an AI approach. In contrast to earlier surveys that focus on fault detection, diagnostics, and prognostics (e.g. Fink et al., 2020), Nguyen et al. (2022) offer a more general guide for implementing AI in PHM, including data acquisition, health indicator construction, and maintenance decision support.

2.2. AI approaches

Atamuradov et al. (2017) classify prognostic methods into three main groups: model-based, data-driven, and hybrid. There are several taxonomies for AI-based prognostic methods. For example, Nguyen et al. (2022) consider four classes of methods based on statistical learning, neural networks, fuzzy systems, and hybrid approaches, respectively. In Table 1, we extend this taxonomy. In Table 1, examples of applications of AI algorithms are given in the review articles that are listed in the footnotes.

In their examination of machine learning (ML) techniques, Chenariyan Nakhaee et al. (2019) distinguish between *shallow* and *deep* learning algorithms based on the level of representation. Shallow techniques use hand-crafted features to learn the mapping between predictive variables and the target variable, whereas deep learning techniques generate features to learn the mapping directly from the data. AI approaches can also be classified based on supervision (Nguyen et al., 2022): unsupervised methods build a mapping between input data and outputs using training data without pre-existing labels, whereas supervised approaches seek a mapping using fully labelled training data. Semi-supervised learning approaches use a mixture of unlabelled and labelled training data. In general, supervised learning approaches achieve high accuracy, but their applications are limited to real-world applications for which consistent data labelling is available (Abid et al., 2021).

2.3. Considerations in the evaluation of AI approaches

The prognostic performance of AI approaches for PHM can be measured with various metrics (Saxena et al., 2008). Ochella and Shafiee (2021) classify PHM performance metrics into conventional, algorithm performance, cost-benefit, and computational performance metrics. Many metrics are limited in that they require the ground truth RUL value, which may not be known. If the purpose of the PHM task is to classify failures, possible performance metrics include accuracy, precision, recall, and specificity (Mahdi and Esztergár-Kiss, 2024). No matter what performance metrics are chosen, no approach can outperform others with every possible test data, as stated by the No Free Lunch theorem (Wolpert, 1996).

Javed et al. (2017) present a scheme for efficient prognostics models that emphasises robustness, reliability, and applicability. In their scheme, *robustness* denotes an approach's resistance to inherent input data variations, while *reliability* signifies an approach's consistency when encountering new data. A reliable prognostics model gives consistent outputs when inputs differ only slightly from learned cases. *Applicability* represents the ability of an approach to be used in industrial settings. The authors note that the robustness, reliability, and applicability of the model depend on the data context and knowledge assumptions, which can be validated by assessing the performance of the model. Validation is especially crucial for building confidence in predicting future outcomes.

Wanner et al. (2020) consider several criteria for making decisions in support of predictive maintenance, highlighting performance, explainability, and effort. These criteria are further categorised based on

Table 1
Taxonomy of AI algorithms.

Algorithm	Brief description	Pros	Cons
Linear models: • Linear regression[1,3] (LR) • Ridge/Lasso regression • Robust LR • Bayesian LR[3]	The expected value of the target variable is a linear function of the explanatory variables and their possible expansions (e.g., polynomials).	• Analytically simple and easy for a non-expert to understand.	• Not useful for non-linear models. • Simpler variants are sensitive to outliers. • Can underfit with small, high-dimensional data.
Markov models[1]: • Hidden Markov models[2] (HMM) • Semi-Markov models (SMM) • Hidden SMM[2]	A Markov model is an infinite sequence of states representing component degradation from new to failed, with transitions depending only on the current state. In SMMs, transitions also depend on the sojourn time in the current state, while HMMs assume degradation is unobservable.	• Appropriate when discrete states can describe degradation. • Analytically simple and easy for a non-expert to understand.	• Can be computationally expensive and require a large number of simulations. • Require the definition of the degradation states and the estimates of the transition probabilities.
Artificial neural networks[1,2,3,4] (ANN): • Convolutional neural networks[1,4,5] • Extreme Learning machines • Radial Basis networks • Recurrent neural networks[2,4,5] (RNN) • Echo State networks (ESN) • Long Short Term memory[4] (LSTM)	ANNs, comprised of neurons connected via weighted connections across input, hidden, and output layers, are trained using data to build regression models that minimise errors between the network's output and the desired target. RNNs and variants like ESN and LSTM include feedback connections, enabling the capture of temporal dependencies and processing of dynamic information. Autoencoders, another type of ANN, are employed for supervised learning of efficient data representations.	• Provide good mappings between input and output data points in many practical PHM instances.	• Need extensive, representative training datasets. • Performance relies on the user's ability to optimise settings (number of neurons, layers, activation functions, etc.). • The operating and training processes are "black boxes", as understanding the built models, except qualitative, is difficult to grasp. • ANNs can have a slow convergence during the training process.
Kernel methods: • Fuzzy Similarity[2] • Support Vector machines[1,2,3,4,6] • Relevance Vector machines • Gaussian Fields	Non-parametric regression methods utilise stored data as a training set. When predicting the value of a new instance, they calculate its distances from or similarities to the available training instances.	• Efficient with small and large datasets. • Can provide real-time analysis and guarantee a good generalisation performance. • Can handle non-linear and complex system modelling. • Models are built directly from the training instances.	• Tuning of parameters strongly affects the performance. • Heavy memory usage for storing all training instances. • Risk of overfitting.
Trees[1,3,4], Bagging and Boosting: • Classification and regression trees (CART) • Random forests[1] (RF) • Adaboost • XGBoost	CARTs are defined by recursively partitioning the input space and defining a local model in each resulting region of the input space. RFs are CARTs aggregated by bagging. Boosting builds predictive models from an ensemble of weak predictive learners.	• Handle mixed continuous and discrete inputs. • Perform automatic variable selection. • Robust to outliers and missing input features. • Fast to fit and scale well to large datasets.	• Although simple CARTs are interpretable, ensembles of trees are not. • Training can be complex.
Model-free approaches: • Prophet • ARMA • ARIMA[1,7] • SARIMA	At the core of these methods is a mathematical model that dissects time series data into distinct components by leveraging past values.	• Build without specific knowledge. • Produce high-quality forecasts for time series data with multiple seasonality with linear or non-linear growth.	• Automatic updates are complex; the modelling process must be reiterated when new data emerges.
Model-based approaches[7]: • Monte Carlo inference (e.g., Particle filtering[2,7]) • Filtering techniques (e.g. Kalman filter[2,7], unscented Kalman filter)	These methods aim to establish physical component models for predicting Remaining useful life (RUL) using mathematical equations that describe the monitored machine's physics and degradation.	• Can outperform model-free approaches when a precise and accurate model is constructed from a thorough physical understanding of the component.	• A physical model of the degradation is required. • Applicability can be restricted by uncertainty due to the assumptions and simplifications of the adopted models.
A mixture of expert models: • Gaussian Mixture[1,7] • Hierarchical mixtures of experts	The mixture contains multiple submodels, each acting as an expert within a particular input space region. A gating function determines which expert to use based on the input values.	• Are flexible. • Easy for a non-expert to understand.	• Need extensive, representative training datasets. • Model parameter estimation can be cumbersome.

[1]Theissler et al. (2021) [2]Kan et al. (2015) [3]Tsui et al. (2015) [4]Nguyen et al. (2022) [5]Lu (2019) [6]Widodo and Yang (2007) [7]Atamuradov et al. (2017).

the risk attitude and the willingness to use AI. The authors conclude that performance is the most important criterion, while explainability and effort hold equal importance. Furthermore, explainability is an important criterion for individuals who are unwilling to use AI. Furthermore, many AI approaches do not provide high performance together with high explainability. For example, deep learning models exhibit high performance but lack transparency and explainability (Monteiro and Reynoso-Meza, 2023). Despite vibrant research on explainable AI (which seeks to enhance the comprehensibility of AI behaviour by providing explanations), explanations are task-specific and depend on the user's abilities and expectations. Therefore, the definitions of interpretability and explainability rely on the domain. According to Zio (2022), attributes sought for in the interpretability include fairness, robustness, causality and quantifiable reliability.

Yet machine learning (ML) algorithms that offer high predictive accuracy may not be adopted in practice (Rana et al., 2014) because companies must also consider criteria such as reliability, overall value, long-term competitiveness of ML algorithms and cost-effectiveness, including setup, operation, and maintenance costs. In addition, Sohn and Kwon (2020) argue that the hedonic aspects of seeking new experiences and curiosity about technology related to AI-based systems are more important than cost–benefit analysis. For example, Dora et al. (2022) summarise the critical success factors for adopting AI in food supply chains. The surveyed experts highlighted key factors such as technology readiness, security, privacy, customer satisfaction, perceived benefits,

Table 2
Summary of categories of the criteria for the development of AI-based solutions.

Category	Criteria
Performance	Prognostic accuracy, robustness, reliability, applicability, computational performance, consistency under data variation (Saxena et al., 2008; Javed et al., 2017)
Explainability and Trust	Model transparency, explainability, interpretability, fairness, causality, quantifiable reliability, user trust (Wanner et al., 2020; Gunning et al., 2019; Zio, 2022; Wanner et al., 2021)
Cost-effectiveness	Setup, operation, and maintenance cost (Rana et al., 2014)
Perceived usefulness and ease of use	User experience, social interaction, customer satisfaction (Liu and Insua, 2020; Sohn and Kwon, 2020; Davis, 1989; Chatterjee et al., 2021)
Organisational Readiness	Curiosity, managerial support, strategic alignment, AI-process fit, technical and managerial capability, organisational culture, financial and personnel resources (Sohn and Kwon, 2020; Jöhnk et al., 2021; Chen et al., 2021)
Data Processing	Data availability, data quality, data accessibility, data flow (Jöhnk et al., 2021)
Environmental Readiness	Technology readiness, security, privacy, regulatory compliance, market uncertainty, partner support, competitive pressure (Dora et al., 2022; Chen et al., 2021)

demand volatility, regulatory compliance, competitor pressure, and information sharing among partners. Likewise, Liu and Insua (2020) focus on social interaction and user experience to facilitate the development of AI-based social robots.

Stated mostly as conceptual models, consumer-oriented theories of technology acceptance offer relevant perspectives also into the adoption of AI approaches, even if there is no consensus on which model is most suited for AI (or for other application areas) (Sohn and Kwon, 2020). Davis (1989) proposes the technology acceptance model (TAM), which explains technology acceptance with perceived usefulness and perceived ease of use. However, the TAM may not be ideal in explaining the adoption of new AI-based intelligent products (Sohn and Kwon, 2020). Chatterjee et al. (2021) extend the TAM by incorporating the technology-organisation-environment (TOE) framework to understand how socio-environmental and technological factors influence the adoption of AI-embedded technology in digital manufacturing. They analyse how organisational factors, such as competency, complexity, readiness, and compatibility, together with external factors like competitive advantage and partner support, impact the perceived usefulness and perceived ease of use.

Chen et al. (2021) integrate the TOE framework with the diffusion of innovation to identify ten success factors impacting AI adoption in the telecom industry at the organisational level. Factors related to the technical context aspect are compatibility, relative advantage, and complexity. The factors associated with organisational capabilities are technical capability, managerial capability, and managerial support. Finally, factors linked to the external environment are government involvement, market uncertainty, competitive pressure, and vendor partnerships. Wanner et al. (2021) employ the unified theory of acceptance and use of technology to measure how trust and system transparency impact the adoption of maintenance decision support systems with AI capabilities.

Based on interviews, Jöhnk et al. (2021) identify 18 factors that contribute to organisational AI readiness to guide the AI adoption process. These factors are categorised into five categories. Among them, the strategic alignment category encompasses AI-business potentials, customer AI readiness, top management support, AI-process fit, and data-driven decision-making. The resources category includes financial budget, personnel, and IT infrastructure. The knowledge category encompasses AI awareness, upskilling, and AI ethics. The culture category includes innovativeness, collaborative work, and change management. Finally, the data category includes data availability, data quality, data accessibility, and data flow. Jöhnk et al. (2021) point out that organisations should continuously assess and develop their AI readiness during AI adoption.

To summarise, the literature is conclusive in suggesting that it is pertinent to consider objectives explicitly in the development of AI solutions. Because there are several criteria, as categorised in Table 2, the question of how these criteria set up development objectives needs to be addressed.

3. Decision process

The proposed framework consists of four development stages and three decision gates. Activities in these development stages produce evidence to guide decisions about (i) which use cases are selected to build corresponding proofs of concept; (ii) which proofs of concept are chosen to underpin detailed implementation plans; and, finally, (iii) which implementation plan will be executed to introduce the AI solution. These main concepts are defined in Table 3, and Fig. 1 illustrates the flow of the process.

Structuring the decision process as a sequence of development stages and decision gates supports the systematic generation of evidence and the evaluation of development objectives. At the outset, tentative ideas concerning the applicability of different AI algorithms for the PHM task using data resources are clarified by specifying use cases. These constitute the initial pool of *candidates*. The technical viability of these use cases is then subjected to exploratory testing, which leads to proofs of concept. Promising proofs of concept provide a basis for detailed implementation plans for the introduction of the AI solution. From this perspective, these three concepts - *use case*, *proof of concept*, and *implementation plan* - are deliverables from consecutive development stages through which the initial vision articulated in the use case is enriched with the details for implementing the AI solution.

The decision gates help to ensure that the principles of resource allocation are clearly stated within the organisation, which helps improve the speed and cost-efficiency of the development process, especially if multiple candidates are developed in parallel. Systematic assessments at the gates also serve to counter premature commitments to technically attractive approaches that may prove infeasible because of a lack of economic viability or problematic partnerships, for example. Typically, these assessments are made by steering groups that have sufficient expertise and act as legitimate "gatekeepers" (Cooper, 1990).

At the decision gates, the following need to be taken into account:

- *Ex post evaluation* at the decision gate, based on the evidence from the previous and also earlier development stages. For example, when choosing use cases at Decision Gate 1, a key criterion is the extent to which the AI algorithm is compatible with data resources.
- *Ex ante evaluation* of the expected impacts of the candidate's development trajectory, assuming that this candidate is chosen as the basis for the AI solution. For example, while some candidates may offer superior technological performance, they may be infeasible if the required investment is excessively high.
- If the results of the ex post and ex ante evaluations are satisfactory, the candidate may be taken to the next stage. In this case, concrete development targets for the next stage are defined, resources are provided, and the planned schedule is stated.

The earlier stages can be revisited in light of the latest evidence. For example, if new and particularly promising AI algorithms become available, development activities based on these may be started, and those based on previously considered algorithms may be terminated.

Table 3
The first three concepts characterise the development context. The next four concepts refer to deliverables from the development stages. *A candidate* is an overarching term that encompasses the deliverables in their respective stages of development.

Concept	Definition
PHM task	The acquisition and transformation of data resources into prognostic information that is employed to support the maintenance of the system.
Data resources	The set of knowledge, information, and data generated by the technical system under consideration.
AI algorithm	A data-driven or hybrid prognostic approach that employs computational procedures to map available data resources into prognostic information used to support maintenance decisions.
Use case	A combination of a PHM task, a data resource, and an AI algorithm; a contextualised description of a specific PHM task in which an AI algorithm can potentially be deployed to contribute to attaining development objectives by harnessing data resources.
Proof of concept	A structured and systematically generated body of evidence supporting or countering the belief that the AI algorithm under consideration is a technically viable candidate that meets the performance requirements specified in the use case description.
Implementation plan	A detailed and scheduled plan, based on an approved proof of concept, which (i) itemises the sequence of development tasks involved in the implementation of the proof of concept, (ii) describes the processes through which the resulting implementation is introduced, operated and maintained, and (iii) includes the assessment of expected economic and strategic impacts of these tasks and processes.
AI solution	A full-scale implementation of an AI algorithm, embedded operationally in the data and IT infrastructure to provide functionalities that meet the performance requirements specified in the use case description.
Candidate	A general term for an AI algorithm that is under development into an AI solution and, depending on the development stage, is evaluated through a use case, a proof of concept, or an implementation plan.

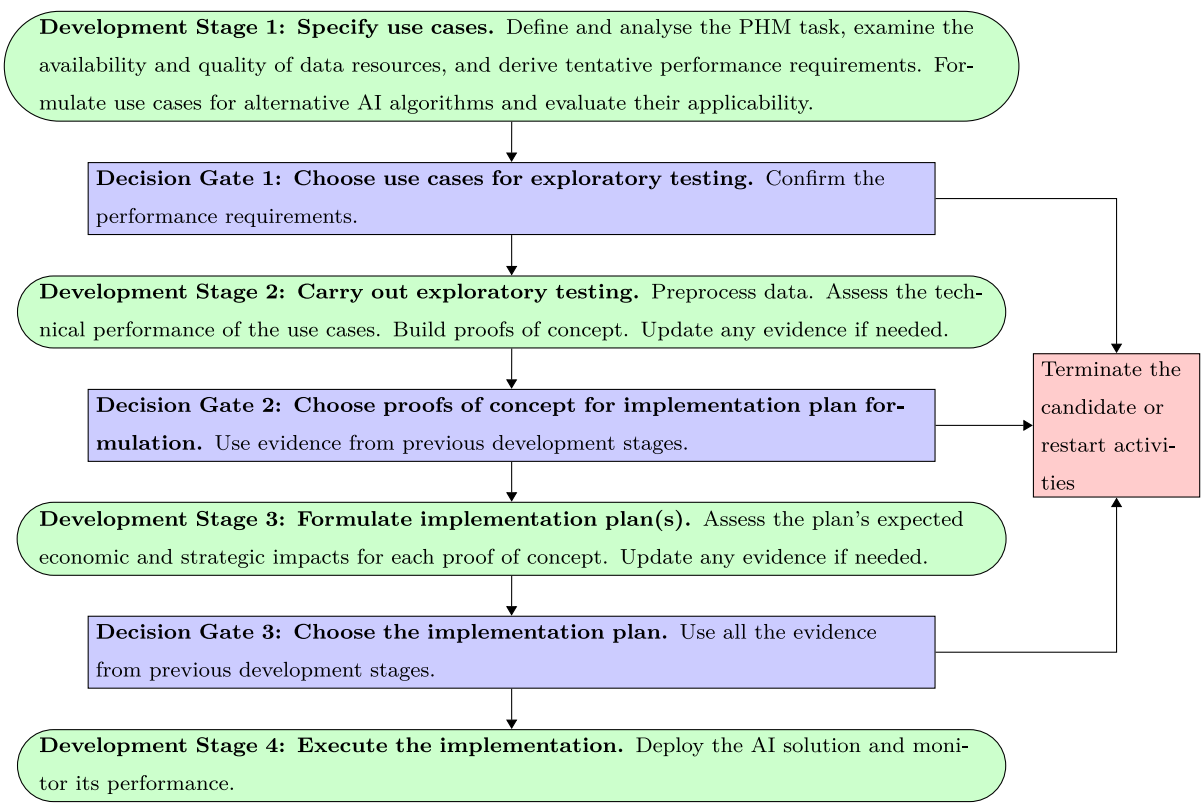

Fig. 1. The candidate proceeds from one stage to the next through go/no-go decisions. In case of a no-go decision, the candidate is either terminated or possibly redirected by reverting to an earlier stage.

3.1. Objectives and criteria

Several objectives can be employed to assess the progress of the development work and the expected performance of the eventual AI solution. These objectives can be operationalised by defining a set of corresponding criteria. Based on the literature review of Section 2.3 and the case study in Section 4, we elaborate on six objectives and derive 17 mutually exhaustive and jointly exclusive criteria for them. Thus, Fig. 2 shows these criteria, which are linked to the objectives: *minimise development costs*, *minimise development time*, *gain economic benefits*, *maximise performance*, *advance capabilities*, and *improve reputation*. However, this hierarchy reflects common but not universal development objectives.

There are m candidates denoted by $X = \{x^1, \dots, x^m\}$. The evaluation of candidate x^j with regard to the ith criterion is x_i^j. A criterion-specific value function $v_i(\cdot)$ maps the evaluations on criterion i to the normalised *score* $v_i^j := v_i(x_i^j) \in [0, 1]$. To account for uncertainties or

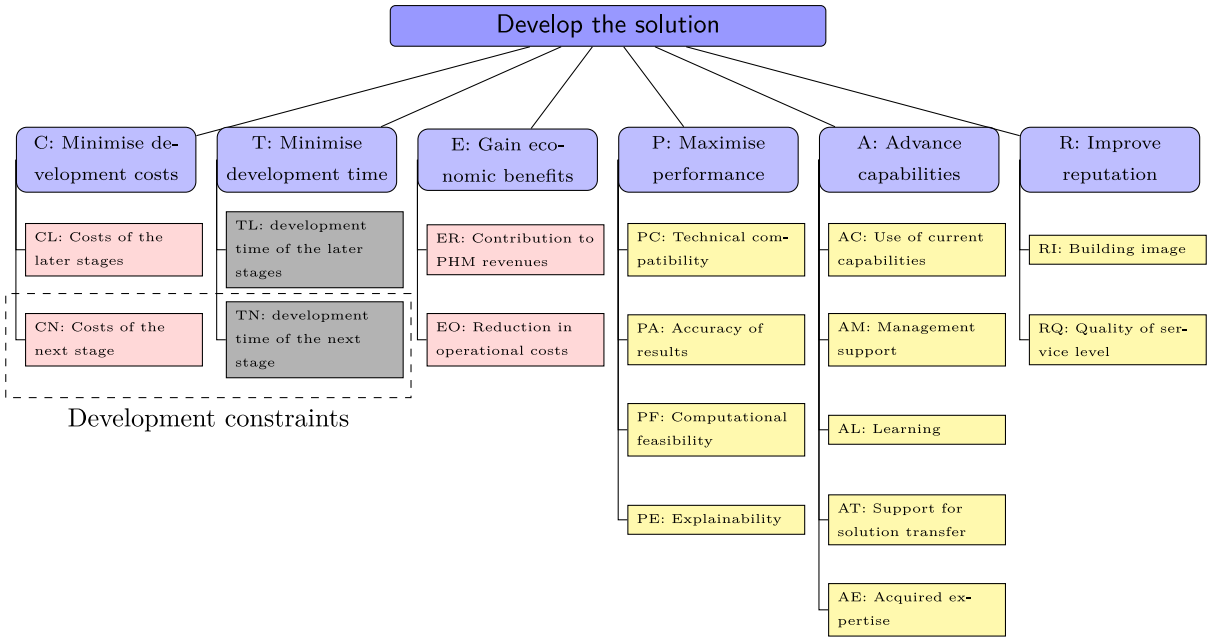

Fig. 2. Hierarchy of objectives and criteria for the evaluation of candidates. The initial letter of each abbreviated criterion refers to the objective. The measurement units of criteria highlighted with pink, grey, and yellow backgrounds correspond to monetary units, time units, and Likert scales, respectively. Criteria CN and TN limit the number of candidates that can be developed in the next stage.

lack of information, this score can be stated as an interval $[\underline{v}_i^j, \overline{v}_i^j]$ where the lower and upper bounds $\underline{v}_i^j$ and $\overline{v}_i^j$ are implied by the respective bounds of the evaluation. Gaining more informative evidence usually allows these score intervals to be tightened. They can be enriched with verbal expert judgements to justify the stated evaluations.

The objectives *minimise development costs* and *minimise development time* refer to the expected monetary and temporal resource requirements. Both of them are operationalised using two criteria. The criterion *costs of next stage* (cf. *costs of later stages*) covers the total expected information technology (IT), personnel, and other direct costs required by the candidate during (cf. after) the next development stage. The criterion *development time of next stage* (cf. *development time of later stages*) refers to the length of time needed to complete the next development stage (cf. later development stages).

The separation of the next stage from the later stages (subsequent stages beyond the next stage) is useful in that the utilisation of these four criteria is treated differently. First, the scores of candidates with respect to criteria *costs and development time of the next stage* define budget and schedule constraints that limit the number of candidates under development. Second, candidates are scored with respect to the criteria representing *costs and development time of the later stages* based on how much costs and time are saved compared to the expected budgets and schedules of later stages. We demonstrate this proposal in Section 4. Thus, the achievement of the two objectives is measured against available stage-specific budgets and schedules.

The development of an AI solution is often motivated by the objective of *gaining economic benefits*. The criterion *contribution to PHM revenues* refers to the additional revenues expected from operating the AI solution during its expected service life or from selling the solution elsewhere. This measurement is based on a total revenue estimate. The criterion *reduction in operational costs* refers to the lower expected operational costs (IT, personnel, maintenance, etc.) compared to current maintenance, based on total expected operational cost savings.

The three objectives with yellow criteria in Fig. 2 refer to the quality of the candidate. The objective *maximise performance* ensures that the candidate meets stated performance requirements, while the objectives *advance capabilities* and *improve reputation* are also relevant to quality. These three objectives, defined in Appendix A, are assessed with a subjective 7-point Likert evaluation $\{1, 2, \ldots, 7\}$ (Likert, 1932), which are converted to normalised scores. The criteria can be evaluated both ex-post based on accumulated evidence and ex-ante based on expectations for the next stages (see the guidelines in Appendix A).

Overall, the evaluation framework helps meet the requirements for structuring fundamental objectives (see, Keeney, 1992). Specifically, the proposed objectives are *Essential* in that for any given objective, there are PHM decision processes in which this objective matters; *Controllable* as the selected candidates can impact the attainment of objectives; *Complete* as the objectives cover a broad range of technical, economic and even broader considerations, such as reputation; *Measureable* and *Operational* as for each objective, the constituent criteria and corresponding measurement scales have been are elaborated; *Decomposable*, evidenced in the itemisation of criteria for each objective; *Nonredundant* as there are no apparent overlaps between the objectives; *Concise*, thanks to the succinct titles; and *Understandable* in that for each objective, a clearly stated non-technical description is provided.

The hierarchy in Fig. 2 offers a general framework for the evaluation of candidates. It can be employed as a template that organisations can tailor to reflect their specific development goals, constraints, and operational context, in order to design a decision process that is aligned with the characteristics of the PHM task to which it is applied. This tailoring may involve, for example, the removal of criteria, the introduction of new ones, or revisions to the scales used for assessing candidates with regard to the criteria.

3.2. Importance of criteria

We distinguish between *"must-meet"* and *"should-meet"* criteria (Grönlund et al., 2010). The requirements that the candidate *has to* fulfil are "must-meet" criteria. If a candidate fails to meet even one, it is terminated. For example, if a performance criterion is evaluated with the 7-point Likert scale, the evaluation must achieve (see statements proposed in Appendix A) the threshold level which represents the minimum acceptable level. Another "must-meet" criterion can be *costs of the next stage* if a budget is set. In contrast, "should-meet" criteria reflect desirable features that inform preference rankings among feasible

options. Even comparatively low evaluations on some "should-meet" criteria may be compensated by high evaluations on others.

From a process perspective, it may be better to first define the complete set of relevant criteria and to classify them into the categories of "must-meet" and "should-meet" criteria later. The distinction between these two types of criteria is based on differentiating between feasibility and desirability: "must-meet" criteria impose hard feasibility constraints on the candidates, whereas "should-meet" criteria facilitate comparisons within the feasible set of candidates. The implications of this classification are non-trivial. Assigning a criterion to the "must-meet" status gives a binding constraint, which may prematurely exclude promising candidates. Conversely, being too permissive about "should-meet" criteria may allow too many candidates to pass the decision gate so that resources are spent on advancing infeasible candidates. Thus, given that this categorisation involves trade-offs, it calls for careful deliberation, particularly when the constraints arising from the availability of resources are tight. For instance, if there are many use cases available as potential proofs of concept, one may provisionally wish to impose relatively stringent "must-meet" requirements at the Decision Gate 1 to ensure that some AI solution can be developed on time. Yet if none can be developed, one may consider possibilities of relaxing some of the earlier "must-meet" criteria.

The categorisation of "must-meet" and "should-meet" criteria depends on the PHM task. For example, *computational feasibility* may be viewed as a "must-meet" criterion because high scores on other criteria cannot compensate for computational infeasibility. In some contexts, *explainability* can be viewed as a "must-meet" criterion because it may be a prerequisite for sufficient trust in the AI solution. However, if black-box solutions are accepted, it may be viewed as a "should-meet" criterion.

Methodologically, while our framework is based on a hierarchical weighting of objectives and their criteria, it does not assume that decision-makers are fully rational in the sense that they would be able to provide complete preference information ex ante, or that they would have to comply with the decision recommendations that are provided. For example, because the uncertainties in the development of an AI solution are partly resolved through the arrival of new evidence, decision-makers are allowed to provide preference information through score and weight intervals, as well as ranking statements about the relative importance of criteria. This incomplete information is then synthesised to explore how the alternatives contribute to the attainment of the objects and corresponding criteria. As illustrated in the case study, even the criteria weights can be adapted to reflect the priorities in the different stages of the process. From this perspective, it is more meaningful to view the proposed decision model as a coherent and comparatively comprehensive evaluation framework that supports mutual learning and exploration (see, e.g., Salo et al., 2021), rather than a separate tool that would seek to mimic the choices of a fully rational individual decision-maker.

Based on the elicited score and weight information, recommendations for selecting candidates are derived with Robust Portfolio Modelling (RPM) (Liesiö et al., 2007, 2008). RPM admits incomplete preference and score information and identifies all feasible portfolios of non-dominated alternatives (in our case, candidates), ensuring that no other feasible portfolio has a higher aggregate value for any scores and weights in the stated information set. This avoids over-reliance on fixed weights and ensures that selected solutions are robust across plausible parameter values. If an alternative is in all non-dominated portfolios, it should be selected; and if it is not contained in any of them, it should be rejected. Otherwise, the selected portfolio will be dominated. Details on computing these recommendations are given in Appendix B.

The rationale for using RPM is that after each decision gate, the development activities will be continued with a portfolio of candidates, and that this portfolio needs to be feasible in the sense that resource constraints are not violated. Conversely, in the presence of incomplete information, an attempt to identify the "best" candidate would be problematic because it would be based on the use of numerical parameters from the feasible information sets, rather than considering these sets as a whole. In particular, the performance of a candidate which would be potentially optimal for some feasible parameters could have a very low overall value for some other parameters. In this setting, RPM accounts for the presence of incomplete information, identifies a set of "good" (i.e., non-dominated) portfolios that satisfy the constraints, and thus facilitates transparent decision-making.

Typically, there is a relatively limited number of use cases at the start of the development process. In addition, the presence of "must-meet" criteria, the characteristics of the PHM task, and the available resources can limit the number of candidates under consideration. Taken together, the number of viable candidates is therefore likely to be modest, which makes the computation of non-dominated portfolios with RPM a tractable problem that can be solved in minutes on a standard computer (Liesiö et al., 2008). However, generating the necessary input for the RPM may take time. First, the candidates' scores are guided by the evaluation and testing activities in the development stages. Second, decision-makers must be consulted to obtain weight intervals, which also takes time. In comparison with how long these two activities typically take, the time required to carry out RPM computations remains very limited.

3.3. Development Stage 1: Specify use cases

The process starts by specifying the use cases. First, the PHM task is studied to characterise the required prognostic information, technical requirements for the AI solution, and the planned scale of operation. For performance-related criteria (e.g., accuracy of results, computational feasibility), "must-meet" minimum performance levels may be specified. For example, a run-time threshold may be imposed if the PHM task must be completed with low latency.

Second, data resources are systematically analysed. Their quality can limit the applicability and performance of AI algorithms (Nguyen et al., 2022). Usual concerns include data cleanliness, acquisition period, number of signals, presence of domain-specific signals, consistency of labels, and access to data about anomalies or failures. For example, supervised AI algorithms should be avoided if data labels are not consistently recorded.

Third, the use case description is completed by specifying how an AI algorithm transforms available data resources into prognostic information. Different use case descriptions can be formulated by pairing the characterisation of the PHM task and data resources with various algorithms (see Table 1). The resulting use cases constitute the initial pool of candidates.

3.4. Decision Gate 1: Choose use cases for exploratory testing

At Decision Gate 1, use cases are evaluated based on performance information about their scores and preference information about the importance of criteria. Based on this information, non-dominated use case portfolios are computed (see Appendix B for details) to provide recommendations. Here, evaluations regarding the costs and time requirements for the next stage are treated as feasibility constraints that influence how many use cases can progress to exploratory testing.

As a rule, only use cases that meet every "must-meet" criterion should be accepted. Among these, use cases that are included in every non-dominated portfolio should be selected for further development, whilst the development of those that are not contained in any non-dominated portfolio may be terminated. Choices among use cases that are in some but not all non-dominated portfolios can be guided by questions such as: (i) Are the use cases compatible with the data resources? (ii) Do they have lower scores in every criterion than some other use case? (iii) Will they receive higher criteria scores when developed further?

In addition to choosing use cases, threshold levels for the required performance, including *computational feasibility* and *accuracy of results*, can be stated towards the development of proofs of concept. In addition, resources can be provided together with a development schedule, guided by the evaluations of the selected use cases.

3.5. Development Stage 2: Carry out exploratory testing

In Development State 2, the use cases are developed through exploratory testing. If the AI algorithm related to a use case is not specific, testing can be time-consuming. Different variants of the algorithm (e.g., linear regression has variants such as Ridge, Lasso, Robust, see Table 1) may need to be implemented for testing, and hyperparameters may require optimisation to maximise performance when developing use cases based on neural networks and kernel methods. If performance is estimated in a lab environment using mock-up systems, expert judgements may be needed to evaluate the robustness of use cases, especially when evaluating the *accuracy of results* criterion (see, e.g., Khan et al., 2021).

Testing gives indications of the contribution of the use case to the objective *advance capabilities*. For example, improving the algorithm produces evidence on the *learning* criterion. Estimates of the expected costs and development time for the next and later stages can be re-evaluated and updated. For each use case, this development stage helps formulate an associated proof of concept.

3.6. Decision Gate 2: Choose proofs of concept for the development of implementation plans

At Decision Gate 2, promising proofs of concept are chosen to formulate implementation plans. Preference information can be revisited to update the feasible weights, while evaluations of costs and development time of the next stage appear in feasibility constraints. Moreover, non-dominated portfolios of proofs of concept can be computed from the updated information set of scores and weights (see Appendix B for details).

Only proofs of concept that meet all "must-meet" criteria should be chosen for formulating an implementation plan. Strong support is given to those proofs of concept that are included in every non-dominated portfolio, while those that are not in any non-dominated portfolio should be terminated. Guiding questions include the following: Is it plausible that the proof of concept will have higher criteria scores than others later on? Again, narrower evaluation intervals or more specific weights are likely to lead to fewer non-dominated portfolios that provide more conclusive recommendations.

If there are enough resources, several proofs of concept can be advanced to formulate corresponding implementation plans, whereby the estimation of required resources can be guided by earlier statements about the expected cost and development time. If no proof of concept appears attractive, one may reassess the situation and consider re-examining previously terminated use cases.

3.7. Development Stage 3: Formulate implementation plans

In Development Stage 3, a detailed implementation plan is formulated for each chosen proof of concept. This plan itemises the requisite implementation activities as well as the processes through which the AI solution will be introduced, operated, and maintained. In addition, the expected economic and strategic impacts of the implementation plans are evaluated to assess the costs and duration of the next stage. Because there are no later development stages, the criteria costs and development time of the later stages need not be evaluated.

The development of an implementation plan generates evidence that is useful for evaluating candidates with respect to the criteria associated with the two objectives *gain economic benefits* and *improve reputation*. To obtain the evaluations of *contribution to PHM revenues* and *reduction of operational costs*, the impacts of implementing the AI solution based on the proof of concept may have to be analysed with a maintenance optimisation model (see, e.g., de Jonge and Scarf, 2020), which can be a challenging task itself.

Because the implementation plan can describe additional activities related to the objective *advance capabilities*, the criteria evaluations may need to be updated. A good implementation plan provides a solid basis for evaluating it with respect to the relevant criteria in Fig. 2 (note that because there are no later stages, the criteria representing the costs and development time of later stages are no longer relevant).

3.8. Decision Gate 3: Choose the implementation plan

At Decision Gate 3, a single implementation plan is selected, guided by the RPM analysis of updated preference information and accumulated evidence. With incomplete criteria scores and preferences, there may be no implementation plan that dominates all others. In this case, the criteria may be revisited with the aim of narrowing the score intervals in view of all available evidence. One may also check if there are other AI algorithms that would enable the development of an AI solution on schedule. Even if the analysis may not provide a conclusive decision recommendation, a systematic evaluation of alternative implementation plans provides a stronger foundation for an informed choice than would be the case in the absence of a structured decision process.

3.9. Development Stage 4: Execute the selected implementation plan

Finally, the selected implementation plan is executed for the PHM task on the intended scale of operations. Using the AI solution produces operational evidence of its realised performance, which can be compared with earlier estimates to support organisational learning and process improvement. Deviations may occur, for example, due to *concept drift* (see, e.g., Lu et al., 2018), which refers to the fact that performance may depend on actual operating conditions that were not known during exploratory testing.

4. Illustration of the decision process

4.1. PHM task and data resources

We illustrate the stage-gate process by considering the development of an AI solution for a regional train's toilet door system. A brushless motor, activated by passengers, operates the door. The PHM task consists of (i) detecting anomalous patterns in motor power signals related to increased door resistance caused either by spring degradation or the accumulation of dirt on the door track, and (ii) predicting the toilet door's remaining operational time. When the motor power requested for door movement exceeds a fixed threshold, the door is automatically marked as out of order and switched to manual operation. Consequently, the door system is neither critical to safety nor relevant to train availability because toilet service can still be guaranteed even if the door motor fails. This system provides a PHM task that has been previously studied in the literature (see, e.g. Xu et al., 2022).

The primary concern of the PHM task is the impact on the company's reputation via customer service. Thus, the AI solution must accurately identify door system failures. In terms of performance, *computational feasibility* is a "must-meet" criterion since the AI solution must run on the control board of the toilet system with limited computational power and memory. Thus, referring to the evaluation statements in Appendix A, if the evaluation of candidate x^j with respect to the performance criterion *computational feasibility* (PF) is $x^j_{\mathrm{PF}} < 4$, the candidate should be terminated.

Data resources are constrained by the following considerations:

- **Data distribution differences:** The training and test data can vary, for example, due to the concept drift (see, e.g., Lu et al., 2018) caused by unpredictable changes in working conditions, sensor or component ageing, or alterations in operating conditions.
- **Training data sources:** Initially, training data, including failure trajectories, can be obtained from laboratory experiments on mock-ups subjected to simulated stress conditions in the absence of field data on faults. Subsequently, data will be collected from doors mounted on trains. As a result, the estimated performance of the algorithms may vary between the stages of the decision process.
- **System variations:** The data collected by monitoring door systems can reflect different behaviours between different door systems due to varying operating conditions and maintenance activities. This scenario is typical for fleet systems.

Differences in training and test data distributions can lead to unsatisfactory results when the trained AI algorithm is tested with data from another context.

In practice, the application of the structured state-gate model to develop an AI solution requires that, once the problem structure has been formulated, (i) managers state preferences for the criteria that represent development objectives, (ii) experts provide evaluation statements building on the body of accumulated evidence, and (iii) analysts calculate non-dominated development portfolios with RPM to derive decision recommendations. When multiple experts provide statements, the ensuing assessment intervals that model them can be decided through discussions (Kangaspunta and Salo, 2014). The third author, an expert in the application domain, provided the preference and evaluation statements for the following numerical example.

4.2. Utilisation of criteria

The methodology for the identification of non-dominated development portfolios is presented in Appendix B. It involves the steps to specify feasibility constraints, calculate scores of candidates w.r.t. criteria, express preferences between objectives (i.e., weights of objectives) and criteria related to them (i.e., criteria weights); and then calculate ranges of values $\underline{v}_o^j$ and $\overline{v}_o^j$ from (B.2) and (B.3) and, finally, check for dominance with (B.4).

In this development process, related to the objectives *minimise development costs* and *minimise development time*, the criteria *costs of the next stage* (CN) and *development time of the next stage* (TN) are treated as resource constraints on the number of candidates that advance to the next stage. These feasibility constraints are updated between decision gates based on the most recent evaluations of candidates w.r.t. these two criteria. If these evaluations are intervals rather than point estimates, the upper bounds of these intervals are employed as worst-case evaluations, which lowers the risk of exceeding the budget or experiencing delays.

A development portfolio is feasible if the total cost of selected candidates does not exceed the next stage budget or the available development time for the next stage. For each stage, the budget is 60 k€, and the available development time is 12 person months. In principle, this means that by increasing the size of the development team by employing more people, it is probable that reductions in the development time can be obtained.

The criteria *costs of the later stages* (CL) and *development time of the later stages* (TL) refer to the expected consumption of resources beyond the next stage. These criteria are scored by comparing the stated evaluations with the total budget and available development time for the later stages. Thus, for CL, there is a linear value function between 0 k€ and $N \cdot 60$ k€ ($v_{\mathrm{CL}}(0) = 1$ and $v_{\mathrm{CL}}(N \cdot 60) = 0$) where N is the number of development stages beyond the next stage. For TL, there is a similar value function between 0 months and $N \cdot 12$ months ($v_{\mathrm{TL}}(0) = 1$ and $v_{\mathrm{TL}}(N \cdot 12) = 0$). The scores on these two criteria define the values of candidates w.r.t. objectives *minimise development costs* and *minimise development time* because the costs and development time of the next state are treated as constraints.

The objective *maximise performance* is the main objective of this PHM task. Use case specification generates evidence w.r.t. criteria *technical compatibility*, *computational feasibility*, and *explainability*. After exploratory testing, proofs of concept are evaluated w.r.t. criterion *accuracy of results*. The scores of candidates j on all performance related criteria i are derived from the 7-point Likert evaluations by $v_i(x_i^j) = [x_i^j - 1]/6,\ x_i^j \in \{1, 2, \ldots, 7\}$. The scores on these criteria can be refined when candidates are re-evaluated in later stages.

If exploratory testing is carried out internally, it suffices to evaluate use cases w.r.t *use of current capabilities* and proofs of concept w.r.t this criterion as well as the four criteria *learning*, *support to solution transfer*, *acquired expertise*, and *management support* when more evidence is available. Re-evaluation of implementation plans against these criteria is also necessary.

Once the development of a candidate has progressed to the formulated implementation plan, sufficient evidence should be available to evaluate the implementation plans w.r.t. criteria linked to objectives *gain economic benefits* and *improve reputation*. Both the criteria *contribution to PHM revenues* and *reduction of operational costs* quantify the economic benefits in monetary units. Thus, the scores of an implementation plan w.r.t. these two criteria can be summed together to derive the value of a candidate w.r.t. objective *gain economic benefits* (details in Section 4.5). On the other hand, the scores of an implementation plan w.r.t. criteria *building image* and *quality of the service level*, which link to objective *improve reputation*, are derived from the 7-point Likert evaluations.

4.3. Use case specification and choosing use cases for exploratory testing

The following six use cases are specified by combining the PHM task and data resources with the following AI algorithms (Table 1), respectively: *linear regression (LR)*, *Markov model (MM)*, *neural network (NN)*, *kernel methods (KM)*, *tree models (TR)*, and *model-free approach (MF)*. Throughout Section 4, we refer to the candidates by the names of the AI algorithms. Table 4 summarises the candidates' evaluations with respect to criteria throughout the process.

Column "G1" of Table 4 shows that the use cases are chosen for exploratory testing based on evaluations w.r.t. eight criteria that are linked to four development objectives: *minimise development costs*, *minimise development time*, *maximise performance*, and *advance capabilities*. The choice of criteria is motivated in Section 4.2. Evaluations of use cases w.r.t. *costs of the next stage* and *development time of the next stage* appear in feasibility constraints. For example, for *neural network* the cost and development time are $c^{NN} = 40$ and $t^{NN} = 6$ respectively. The scores of use cases w.r.t. *costs of later stages* and *development time of later stages* model the values on *minimise development costs* and *minimise development time* respectively.

The use cases are evaluated w.r.t. three out of the four criteria linked to the objective *maximise performance* (P). Of these, *computational feasibility* (PF) is the most important, *technical compatibility* (PC) the second most important, and *explainability* (PE) the third most important criterion. In addition, a lower bound of 0.2 is set for weights to ensure that the values of use cases reflect all three criteria. These constraints can be expressed with the notation introduced in Appendix B as $w_{\mathrm{PF}}^{\mathrm{P}} \geq w_{\mathrm{PC}}^{\mathrm{P}} \geq w_{\mathrm{PE}}^{\mathrm{P}} \geq 0.2$. Of the five criteria linked to the objective *advance capabilities* (A), use cases are only evaluated w.r.t. the criterion *use of current capabilities* (AC). Thus, the values of use cases w.r.t. objective A are the scores w.r.t. AC.

Preferences between the four development objectives are stated. As depicted in Section 4.2, *maximise performance* is the most important objective. This is interpreted so that the change from the worst value to the best value in objective *maximise performance* is preferred over similar value changes in other objectives. In addition, a lower bound of

Table 4
Candidates (in columns 5–10) are evaluated w.r.t. the criteria (in rows). Abbreviations of criteria are below the table. The measurement scale for the last 11 criteria is a 7-point Likert scale. Evaluations are performed at the relevant Decision Gates. If a criterion is relevant in Decision Gate 1, 2, or 3, it is marked with an "X" in the respective column "G1", "G2", or "G3". The evaluations obtained during different development stages are separated by "/" and presented as "1/2/3", where "1", "2", and "3" refer to evaluations obtained during use case specification, exploratory testing, and implementation plan formulation, respectively. Evaluations that were not conducted are denoted with "-".

Criterion [unit]	G1	G2	G3	*Linear regression (LR)*	*Kernel method (KM)*	*Tree model (TR)*	*Markov model (MM)*	*Neural Network (NN)*	*Model free (MF)*
CN [k€]	X	X	X	4-6/10-12/15	15-25/25-35/-	15-25/35-40/50	15-25/-/-	20-40/-/-	15-25/-/-
CL [k€]	X	X		15-30/15-20/-	40-80/30-40/-	45-85/45-50/-	25-50/-/-	50-90/-/-	35-50/-/-
TN [mth]	X	X	X	1-2/2-4/5–6	2-4/3-5/-	2-3/3-5/7–8	1-3/-/-	3-6/-/-	2-3/-/-
TL [mth]	X	X		4-10/3-6/-	7-13/4-8/-	7-13/4-8/-	5-12/-/-	8-18/-/-	7-12/-/-
ER [k€]			X	-/-/10–15	–	-/-/10–25	–	–	–
EO [k€]			X	-/-/55–75	–	-/-/55–85	–	–	–
PC	X	X	X	2-5/5/5	3-6/4/-	4-6/5/5	4-5/-/-	3-6/-/-	1-5/-/-
PA		X	X	-/3-4/3–4	-/4-5/-	-/5/5	–	–	–
PF	X	X	X	6-7/6/6	2-4/4/-	3-5/4/4	2/-/-	2-4/-/-	3-4/-/-
PE	X	X	X	7/7/7	4/4/-	4/4/4	6/-/-	3/-/-	4/-/-
AC	X	X	X	6-7/7/6	5-6/5-6/-	3-4/5/4	4-5/-/-	2-4/-/-	3-4/-/-
AL		X	X	-/2/3	-/5/-	-/5-6/5	–	–	–
AT		X	X	-/3-4/3	-/5/-	-/4-5/5	–	–	–
AE		X	X	-/3/3	-/6/-	-/6/5	–	–	–
AM		X	X	-/3-4/4	-/4/-	-/5/6	–	–	–
RI			X	-/-/3	–	-/-/6	–	–	–
RQ			X	-/-/2	–	-/-/5	–	–	–

CN: costs of the next stage, CL: costs of the later stages, TN: development time of the next stage, TL: development time of the later stages, ER: contribution to PHM revenues, EO: reduction of operational costs, PC: technical compatibility, PA: accuracy of results, PF: computational feasibility, PE: explainability, AC: use of current capabilities, AL: learning, AT: support for solution transfer, AE: acquired expertise, AM: management support, RI: building image, RQ: quality of the service level.

0.1 is set for the weights of objectives to ensure that the overall value of the use case reflects all four objectives. With the notation of Appendix B, the weights can be expressed as $w_{\mathrm{P}} \geq w_{\mathrm{o}} \geq 0.1$ for all $o \in \{\mathrm{C, T, A}\}$.

The development of a *Markov model* can be terminated after obtaining the evaluations of use cases since it does not satisfy the "must-meet" criterion of computational feasibility ($x_{\mathrm{PF}}^{MM} = 2 < 4$). From the five remaining use cases, there are three non-dominated portfolios: $P_N = \{\{LR, TR, MF\}, \{LR, KM, MF\}, \{LR, KM, TR\}\}$. *Linear regression (LR)* is included in all of those, and *neural network (NN)* in none of those, meaning that *linear regression* is a core use case that should be chosen for exploratory testing and *neural network* is an exterior use case that can thus be terminated (Liesiö et al., 2007).

Based on the non-dominated portfolios, any two of the three remaining use cases *kernel method (KM)*, *tree model (TR)*, and *model-free approach (MF)* can be chosen. Although the process does not provide a single decision recommendation, we have narrowed the number of use cases, whose selection is still uncertain, from six to three. A detailed discussion of how well these three use cases are expected to perform in the objectives and criteria that have yet to be evaluated can justify which two should advance to testing. In this illustration, the discussions justify choosing *KM* and *TR* for exploratory testing (together with *linear regression*) and terminating *MF*.

4.4. Explorary testing and choosing proofs of concept to implementation plan formulation

Use cases *linear regression*, *kernel method*, and *tree model* are elaborated and submitted to exploratory testing to produce proofs of concept with evidence that is evaluated with respect to all 13 criteria linked to four development objectives: *minimise development costs* (C), *progress quickly* (Q), *maximise performance* (P), and *advance capabilities* (A) (see column G2 of Table 4). The results of these evaluations are presented in columns 5–7 of Table 4. Computationally feasible implementations of a *kernel method (KM)* and *tree model (TR)* were obtained as their evaluation w.r.t. *computational feasibility* is 4. Thus, each proof of concept fulfils this "must-meet" criterion. Evaluations are normalised to scores with value functions described in Section 4.2. Overall, the score intervals are tightened after exploratory testing.

The feasibility constraints are updated as the evaluation of proofs of concept w.r.t. *costs and development time of the next stage* has changed. Similarly, the values of the proofs of concept w.r.t. objectives *minimise development costs* and *minimise the development time* are redetermined as the scores w.r.t. criteria *costs and development time of the later stages* are updated.

Of the four criteria associated with the objective *maximise performance* (P), *accuracy of results* (PA) is now the most important and *explainability* (PE) the second most important criterion. In addition, a lower bound of 0.1 is set for weights to ensure that the values of proofs of concept reflect all four criteria. Mathematically, the weights can be expressed as $w_{\mathrm{PA}}^{\mathrm{P}} \geq w_{\mathrm{PE}}^{\mathrm{P}} \geq w_i^{\mathrm{P}} \geq 0.1$ for both $i \in \{\mathrm{PC, PF}\}$. Of the five criteria linked to the objective *advance capabilities* (A), *use of current capabilities* (AC) is the most important and *management support* (AM) is the second most important criterion. Setting a lower bound of 0.05 on weights ensures that all five criteria are represented in the values of proofs of concept. Mathematically, the weights can be expressed as $w_{\mathrm{AC}}^{\mathrm{A}} \geq w_{\mathrm{AM}}^{\mathrm{A}} \geq w_{\mathrm{i}}^{\mathrm{A}} \geq 0.05$ for all $i \in \{\mathrm{AL, AT, AE}\}$.

Preferences between the four development objectives are updated. *Maximise performance* remains the most important objective. In addition, a lower bound of 0.15 is set for the weights of objectives to ensure that the overall value of the proof of concept reflects all four objectives and to prevent a disproportionately large weight for objective *maximise performance*. With the notation of Appendix B these can be expressed as $w_{\mathrm{P}} \geq w_{\mathrm{o}} \geq 0.15$ for all $o \in \{\mathrm{C, T, A}\}$.

Proofs of concepts include *linear regression (LR)*, *kernel method (KM)* and *tree model (TR)*. Due to the budget, choosing both *KM* and *TR* is infeasible. However, the remaining pairs of proofs of concepts are non-dominated, $P_N = \{\{LR, KM\}, \{LR, TR\}\}$. *LR* is a core proof of concept that should be chosen for implementation plan formulation. Furthermore, a detailed discussion of how well *KM* and *TR* are expected to perform in the criteria linked to objectives *gain economic benefits* and *improve reputation* can help justify the choice between the two. In this illustration, the discussions support choosing *LR* and *TR* for the implementation plan formulation and terminating *KM*.

4.5. Implementation plan formulation and choosing the plan to implement

The implementation plans for the proofs of concept *linear regression* and *tree model* are formalised and assessed as described in Section 3.7. The results of the evaluations are in columns 5 and 7 of Table 4.

Corresponding scores are obtained with value functions described in Section 4.2. Notably, after determining the evaluation intervals of implementation plans w.r.t. criteria *contribution to PHM revenues* (ER) and *reduction of operational costs* (EO), the evaluations of implementation plans w.r.t. ER and EO are added to obtain values w.r.t. objective *gain economic benefits* with a value function that linearly increases between 65 k€ and 110 k€: for implementation plan j, $v_{\mathrm{E}}^{j} = \frac{x_{\mathrm{ER}}^{j} + x_{\mathrm{EO}}^{j} - 65}{110-65}$.

The evaluations of implementation plans w.r.t. *costs and development time of the next stage* update the feasibility constraints. As implementation plans are not evaluated w.r.t. *costs and time of later stages*, the overall value of the implementation plan does not depend on the values w.r.t. *minimise development costs* and *minimise development time* and thus $w_{\mathrm{C}} = w_{\mathrm{T}} = 0$. Preferences between criteria linked to objective *maximise performance* (P) remain unchanged. In addition, the evaluations of implementation plans w.r.t. these criteria match the evaluations obtained during building proofs of concepts. Thus, the values of implementation plans w.r.t. *maximise performance* do not change.

For the objective *advance capabilities* (A), *learning* (AL) is now ranked the most important and *management support* (AM) is the second most important. In addition, a lower bound of 0.1 is set for weights to ensure that the values reflect all five criteria. Mathematically, the weights can be expressed as $w_{\mathrm{AL}}^{\mathrm{A}} \geq w_{\mathrm{AM}}^{\mathrm{A}} \geq w_{\mathrm{i}}^{\mathrm{A}} \geq 0.1$ for all $i \in \{\mathrm{AC}, \mathrm{AT}, \mathrm{AE}\}$.

Regarding the objective *improve reputation* (R), the score change from the worst level to the best level in *quality of service level* (RQ) is at least 1.5 times more valuable and at most 2.5 times more valuable than the score change from the worst level to the best level in *building image* (RI). Mathematically, the weights are $\frac{5}{2} w_{\mathrm{RI}}^{\mathrm{R}} \geq w_{\mathrm{RQ}}^{\mathrm{R}} \geq \frac{3}{2} w_{\mathrm{RI}}^{\mathrm{R}}$.

As mentioned above, objectives *minimise development costs* and *minimise development time* impact the decision as feasibility constraints. Preferences between the other four development objectives are stated. *Maximise performance* is the most important, and *gain economic benefits* is the second most important objective. A lower bound of 0.15 is set for the weights of objectives to ensure that the overall value of the implementation plan reflects the four objectives. With the notation of Appendix B these can be expressed as $w_{\mathrm{P}} \geq w_{\mathrm{E}} \geq w_{\mathrm{o}} \geq 0.15$ for all $o \in \{\mathrm{A}, \mathrm{R}\}$.

The budget covers the execution of a single implementation plan. With these scores and preferences, both implementation plans are non-dominated. However, *tree model* can obtain a higher overall value than *linear regression* in this information set. Furthermore, one can conduct sensitivity analyses to explore which changes in the information set would make one implementation plan dominate the other. The decision-maker can then determine whether these changes are acceptable to justify choosing one implementation plan over another.

The stage-gate process supports the decision-making up to this point. In Development Stage 4, the chosen AI solution is implemented. This can create additional evidence that helps to carry out future AI solution development processes.

5. Discussion

The criteria in the proposed decision process span the range from technical performance to economic benefits and even impacts on the organisation's reputation. Although providing reliable estimates for the latter, in particular, may be challenging, doing so helps mitigate the risk of focusing solely on technically promising candidates that may not yield significant broader benefits. Moreover, the consideration of many criteria supports decision quality (Keisler, 2011) and helps justify why one candidate may merit more attention than another despite superior technological performance. Consequently, in Section 4, several non-dominated portfolios were identified at each decision gate. The process also offers a comprehensive framework for decision-making and organisational learning. If systematic biases of over-optimism in the stated evaluations are observed, these can be mitigated through calibration, especially in large organisations that produce more data to support calibration. For example, in Section 4, the value of objective *gain economic benefits* was calibrated based on the evaluations of the candidates with regard to the criteria linked to this objective.

Evidence will be generated only for those candidates that have been approved in the early decision stages of the process, even if these candidates may be rejected later. In hindsight, it would have been better to terminate these candidates earlier. In contrast, there will be little evidence of candidates that have been terminated early on, even if the development of such candidates could have led to a successful AI solution. The empty cells of Table 4 reflect this. Thus, the accumulated evidence will be asymmetric in terms of false positives (i.e., candidates on which a 'go' decision was made, even if it would have been better to terminate development) and false negatives (i.e., candidates that were terminated with a 'no-go' decision, even if their later development would have led to successful implementation).

The risks caused by false negatives are arguably higher in the early stages of the decision process. That is, if the activities in the early development stages consume fewer resources than the later ones, one should not be too stringent in making 'no-go' decisions to ensure that sufficiently many candidates are retained, thereby increasing the probability that at least one candidate will lead to a successful AI solution (e.g. Vilkkumaa et al., 2015). In the illustrative example, more candidates would have passed Decision Gate 1 if the budget had been larger and the available development time had been longer. Even the urgency of the development process may have to be accounted for. In particular, if the AI solution must be developed under a tight schedule and there are sufficient development resources, several candidates may be advanced in parallel. Conversely, adopting a sequential approach would entail the risk that the sole candidate under development will fail in later stages, necessitating the initiation of the process from another use case specification.

Most of the time and effort will likely be spent on the activities that take place during the development stages, which precede decision gates. These activities would depend on the PHM task and the chosen algorithm. Broadly seen, though, key concerns in the specification of use cases are likely to focus on topics such as data quality (Chenariyan Nakhaee et al., 2019) and the initial choice of AI algorithms for the description of use cases (Nguyen et al., 2022). During exploratory testing, targets for the generation of evidence are likely to pertain to performance evaluation (Saxena et al., 2008), robustness analysis (Khan et al., 2021), and concept drift (Lu et al., 2018). The formulation of implementation plans includes the evaluation of economic benefits (de Jonge and Scarf, 2020) and the development of implementation guidelines as well (Nguyen et al., 2022). After executing the implementation plan, it may be instructive to revisit documentation from the stage of exploratory testing to assess if the AI solution performs up to earlier estimates. The activities of the development stages were not discussed in the illustrative example.

When establishing decision-making groups for the decision gates, the organisation's management practices must be considered. It is also essential to ensure that sufficient expertise is brought in so that decisions reflect the organisation's values and goals. Additionally, the decisions should build on relevant sources of expertise across all criteria, including technological, economic, and reputational factors, to derive weights for objectives and criteria that model the preferences. Having some overlap between decision-making groups can facilitate information sharing and promote consistency in the interpretation of criteria. Hence, it may be worthwhile to appoint a designated steering group that meets regularly to review the progress of all candidates.

In implementing AI solutions, organisations often have two choices that are not mutually exclusive: to build their expertise or to rely on external experts. If an organisation lacks in-house expertise, bringing in external experts can be the faster route, especially if the development context is clear and the desired AI solution resembles well-known solutions. In this case, effective communication is needed to ensure that the external experts provide sufficient information about the candidate

throughout the decision process so that the resulting AI solution is guaranteed to meet the initial use case requirements. However, suppose the context involves major ambiguities that cannot be assessed at the outset. In that case, it may be expedient to build in-house expertise, especially if the AI solution must meet unique requirements beyond the experience of external experts.

The question of what PHM tasks AI solutions should be developed for also merits attention. Here, some guidance can be provided by recognising that any such solution is essentially a transformation of data inputs to output signals that call for requisite control actions. Accordingly, PHM tasks for which such an input–output transformation can be unequivocally stated are more likely to be amenable. Conversely, if it is difficult to specify this transformation (e.g., due to uncertainties in data availability or parallel uses of the signal across many parts of the system), the performance of the system is harder to validate. This is particularly the case if the adequacy of controls depends not only on the data but also on the internal condition of the system (which may deteriorate over time, necessitating possible changes in controls) or the external operating conditions (e.g., weather). Thus, when there are many such dependencies, it is more difficult to develop a use case description that spans all the eventual internal and external conditions to enable subsequent evaluation and approval.

Our framework can be adapted and extended for many PHM tasks through modifications. First, the objectives and criteria can be revised for different PHM contexts (such as the development of hybrid methods for failure detection and diagnostics) while retaining the hierarchical structure (see, e.g., Sohn and Kwon, 2020; Wanner et al., 2021; Chatterjee et al., 2021; Chen et al., 2021; Dora et al., 2022). Second, alternative approaches could be employed to elicit and synthesise expert judgements by drawing on multicriteria methods for group decision-making (e.g. Kilgour and Eden, 2010; Salo et al., 2021). Third, the protocols at the decision gates could be extended to incorporate decision rules to ensure, for instance, that there is a sufficient consensus on the candidates that are taken forward (see e.g., Liesiö et al., 2021). Finally, analogous decision models could be applied to assess the performance of the AI solution when it has been taken into operational use.

The following recommendations are suggested for practical adoption. First, to address the PHM task-specific selection and weighting of criteria, it may be useful to start the process with a workshop in which stakeholders express their preferences by setting priorities and by discussing trade-offs. Second, the process can be scaled to align with the available resources in the organisation. For example, small start-ups may benefit from a relatively straightforward qualitative assessment of criteria in view of developing a single AI solution. In contrast, large companies with numerous PHM tasks and candidates under development may adapt the process by keeping track of and sharing results from completed development stages across multiple PHM tasks, given that the development of solutions for these tasks may call for the use of resources (e.g., expertise of IT teams) that are shared among many PHM tasks.

Still, the proposed decision process has limitations. First, the proposed evaluation framework assumes that the objectives and evaluation criteria can be agreed on, elaborated, and operationalised in advance. Yet, in the face of high uncertainties related to novel technical solutions, this can be challenging, especially for criteria that relate to intangible societal or ethical impacts. Second, the high degree of uncertainty may imply that it is hard to elicit well-founded judgements or that the evaluators may provide inconsistent evaluations across candidates. Furthermore, if the candidates do not differ with regard to certain criteria, then evaluating them based on these criteria may consume time without distinguishing between them. For example, if an organisation already has a very strong AI brand, then evaluating candidates with regard to criteria associated with the objective *improve reputation* is unlikely to make much difference. Third, while the additive structure of the decision model helps understand how the value of the candidate is formed, it may be an oversimplification if, for example, there are synergies such that the contribution to the overall value by two high-value scores is greater than the weighted sum of these scores. Fourth, this framework presumes that there are sufficient organisational resources to apply it. In smaller companies or situations where development must be completed under stringent time constraints, such resources may not be available. Finally, while the framework makes the decision process more transparent and manageable, it does not guarantee eventual implementation success, which also depends on external factors such as data compatibility and integration capabilities.

6. Conclusion

In this paper, we have proposed a stage-gate process for guiding the selection of AI solutions for effective smart maintenance, which synthesises data resources into usable prognostic information. A key concept in this decision process is the *use case*, defined as a combination of a PHM task, an AI algorithm, and a data resource. Specifically, the use case specifies how the AI algorithm transforms available data resources into prognostic information to accomplish the PHM task for which an AI solution is sought. Because multiple viable AI algorithms may be considered for the PHM task, the initial pool of candidates encompasses a range of use cases.

The evidence accumulates iteratively. Once the use cases have been described, the development of an AI solution progresses through three subsequent development stages to generate a growing body of evidence through the activities of carrying out exploratory testing, building proofs of concept, and formulating implementation plans. After each development stage, multiple criteria associated with six main objectives are applied to evaluate the accumulated evidence to support decisions regarding the candidates under development. Specifically, these decision recommendations promote candidates that minimise development costs and development time, maximise performance, advance capabilities, gain economic benefits, and improve reputation. Much of the evidence considered at the decision gates following the development stages is evaluated by eliciting expert ex-ante and ex-post judgements. Technically, the evaluation of the candidates may be expressed as score intervals to reflect uncertainties in such evaluations. Expert judgement is relied on extensively in assessing the performance of candidates with regard to qualitative criteria.

In this study, it is assumed that the value model is additive and that decision-makers can operate with bounded rationality, holding coherent but possibly incomplete preferences. On this basis, Robust Portfolio Modelling (RPM) has been applied to find dominance relations between portfolios. The approach remains computationally feasible in the PHM context, because the number of candidates in the portfolios is typically modest, as, for instance, the presence of "must-meet" criteria may restrict the number of candidates under consideration. Importantly, adopting a portfolio-level perspective not only supports decision-making under resource constraints but also enables the parallel development of multiple candidates, which mitigates the risks of premature commitment to suboptimal solutions.

In addition to guiding the development, the structured process serves broader organisational purposes. First, the alignment of evaluation criteria with development objectives helps ensure that development activities are transparently connected to the organisation's priorities. Second, the explicit articulation of ex-post and ex-ante evaluations promotes transparency by making the grounds for decision-making visible. Third, all candidates are treated equally in the process: all use cases, proofs of concept, and implementation plans are subjected to the same systematic assessment. Fourth, the use of predefined stages and decision gates provides decision-makers and development teams with a predictable roadmap of how and when evidence will be assessed. Fifth, the requirement to state and justify the evaluations at each decision gate promotes accountability for decisions concerning the choice of selected candidates. Finally, applying the same stage-gate process across PHM tasks facilitates organisational learning and even continuous improvement, given that experiences from different tasks can be shared.

CRediT authorship contribution statement

Jussi Leppinen: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Ahti Salo:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **Michele Compare:** Writing – review & editing, Writing – original draft, Project administration, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Evaluation statements

This appendix presents a representative sample of verbal statements (in bullets) for the eleven primarily qualitative criteria that can be used to assess candidates on a 7-point Likert scale (1 = strongly disagree, 2 = disagree, 3 = somewhat disagree, 4 = neither agree nor disagree, 5 = somewhat agree, 6 = agree, 7 = strongly agree). Additional statements can be derived, for example, from Sohn and Kwon (2020), Wanner et al. (2021), Chatterjee et al. (2021), Chen et al. (2021), and Dora et al. (2022). If several statements from these lists are employed, the corresponding score interval can be derived, for example, from the minimum and maximum scores of different statements; it is also possible to arrive at the final criterion score with the Delphi method (Linstone et al., 1975). Using these criteria requires different kinds of expertise, and hence, different sets of experts may need to be consulted.

Criteria linked to objective maximise performance

Technical compatibility refers to the extent to which the candidate is technically compatible with the data resources that are available for the generation of prognostic information.

- The requirements of the candidate match the available data resources.
- The development of a candidate can utilise available expert knowledge of the PHM task (see Nguyen et al., 2022).

Accuracy of results refers to the extent to which the diagnostic information provided by the AI solution is accurate enough for the successful completion of the PHM task.

- The root mean squared error satisfies the performance requirement.
- The number of false positives in prognostic information satisfies the performance requirement.
- The number of false negatives in prognostic information satisfies the performance requirement.
- The reliability of the prognostic information satisfies the performance requirement.
- The robustness of the prognostic information satisfies the performance requirement.

Computational feasibility refers to the extent to which the candidate is compatible with the computational requirements for the successful completion of the PHM task.

- The run time of the candidate is below the threshold derived from the requirements of the PHM task.
- The memory requirements of the candidate are below the threshold derived from the requirements of the PHM task.

Explainability refers to the extent to which the results of the AI solution can be logically explained and understood by describing the algorithm and the data resources from which the results are generated.

- The working principle of the candidate is understood.
- The results obtained with the candidate can be interpreted with the data resources.
- The process of using the candidate is easily understandable (adapted from Chatterjee et al., 2021).
- It would be understandable why this candidate provided the prognostic information it did (adapted from Wanner et al., 2021).
- I would easily trust this candidate (adapted from Wanner et al., 2021).

Criteria linked to objective advance capabilities

Use of current capabilities refers to the extent to which the organisation possesses and is able to exploit relevant technical capabilities in developing the AI solution:

- The organisation has all the necessary technical resources to develop the candidate into an AI solution in the PHM task (adapted from Chatterjee et al., 2021).
- In the organisation, there are experts in the AI algorithm under consideration (adapted from Chatterjee et al., 2021).
- The development of a candidate can utilise current IT infrastructure.

Management support refers to the extent to which management understands the potential of AI for completing the PHM task and is prepared to make commitments to the development an AI-based solution.

- The management supports the development of the candidate 1u (adapted from Chen et al., 2021).
- The management is willing to take risks in developing the candidate (adapted from Chen et al., 2021).
- The management understands how the candidate can contribute to economic benefits (adapted from Chen et al., 2021).

Learning refers to the extent to which the development of the AI-based solution contributes to the strengthening of related skills and can be adopted from the viewpoint of not requiring much training.

- It is easy to learn to use the candidate (adapted from Chatterjee et al., 2021).
- The development of the candidate would increase skills related to AI in the organisation.

Support for solution transfer refers to the extent to which the development of the candidate is useful for the development of further AI-based solutions for other PHM tasks in the organisation.

- The candidate helps solve other PHM tasks.
- The implementation of the candidate advances the compatibility of the IT infrastructure towards other AI algorithms.

Acquired expertise refers to the extent to which the development of the candidate increases the expertise related to the PHM task and the available data resources.

- The development of the candidate increases the expertise of the PHM task (adapted from Chatterjee et al., 2021).
- The development of the candidate increases the knowledge of the system maintained.
- The development of the candidate increases the knowledge related to the data resources.

Criteria linked to objective improve reputation

Building image refers to the extent to which the development of an AI solution contributes to the enhancement of the public image of the organisation.

- Implementation of a candidate can promote a positive image of the organisation.
- Implementation of a candidate can improve the image of the organisation in relation to competitors (adapted from Chen et al., 2021).

Quality of the service level refers to the extent to which the adoption of the AI solution improves the quality of service that is offered by using it to complete the PHM task.

- AI solution is advantageous for the organisation (adapted from Chatterjee et al., 2021).
- AI solution can improve customer service (adapted from Chen et al., 2021).
- AI solution would perform maintenance decision support very well (adapted from Wanner et al., 2021).

Appendix B. Robust portfolio modelling

At each decision gate, portfolios of candidates $x^j \in X = \{x^1, \dots, x^m\}$ are screened with Robust Portfolio Modelling (RPM) by evaluating the candidates with the relevant subset of n criteria (Liesiö et al., 2007, 2008). In the hierarchical framework proposed in Fig. 2, these criteria are structured under six higher-level objectives, denoted by $o \in O$ so that $N_o \subset \{1, \dots, n\}$ contains the criteria associated with the objective o.

Preferences about the relative importance of objectives and criteria are represented by weights $w_o, o \in O$ and $w_i^o, i \in N_o, o \in O$, respectively. The *value* of the candidate x^j with regard to objective o is $v_o^j = \sum_{i \in N_o} w_i^o v_i^j$ where $v_i^j := v_i(x_i^j) \in [0,1]$ is the *score* of candidate x^j, that is obtained from evaluation x_i^j with criterion-specific value function $v_i(\cdot)$. The lower and upper bounds of these scores, derived, for example, from the intervals in Table 4, are denoted by $\underline{v}_i^j$ and $\overline{v}_i^j$ respectively.

The weights $w^o = (w_i^o)_{i \in N_o}$ are in set S_o which is a subset of unconstrained weights,

$$w^o \in S_o \subseteq \left\{ w^o \in \mathbb{R}^{|N_o|} | w_i^o \geq 0, \sum_{i \in N_o} w_i^o = 1. \right\} \tag{B.1}$$

This set S_o models the relative importance of criteria linked to objective $o \in O$. If $w_i^o = 0$, then the score v_i^j does not impact value v_o^j.

Notably, $w_i^o = 0$ for every $i \notin N_o$. The weights can be presented in a matrix $w^c \in \mathbb{R}^{n \times |O|}$ where $[w^c]_{io} = w_i^o$. The set of feasible criteria weights is denoted with C_w. Now, values v_o^j can be presented in a matrix $vw^c \in \mathbb{R}^{m \times |O|}$ where $[vw^c]_{jo} = v_o^j$ when $v \in S_v$ and $w^c \in C_w$. These values are updated if the scores v or weights w^c are updated when the development progresses. Nevertheless, values v_o^j are bounded, and their lower and upper bounds are obtained from

$$\underline{v}_o^j = \min_{w^o \in S_o} \sum_{i \in N_o} w_i^o \underline{v}_i^j \quad \text{and} \tag{B.2}$$

$$\overline{v}_o^j = \max_{w^o \in S_o} \sum_{i \in N_o} w_i^o \overline{v}_i^j. \tag{B.3}$$

The *overall value* of candidate x^j is given by an additive value function $V(x^j) = \sum_{o \in O} w_o v_o^j$. The weights $w = (w_o)_{o \in O}$ are normalised so that

$$w \in O_w \subseteq \left\{ w \in \mathbb{R}^{|O|} | w_o \geq 0, \sum_{o \in O} w_o = 1 \right\}$$

where the set O_w includes feasible weights of objectives that model the relative importance of objectives. The information set S is a Cartesian product of feasible scores, feasible criteria weights, and feasible weights of objectives: $S = S_v \times C_w \times O_w$. The overall value of the candidate varies in the information set.

A development portfolio $p \subseteq X$ is a subset of candidates. It is denoted by a binary vector $z(p) \in \{0,1\}^{m \times 1}$ such that $z_j(p) = 1$ if $x^j \in p$ and $z_j(p) = 0$ if $x^j \notin p$. The overall value of a portfolio is

$$V(p, v, w^c, w) := \sum_{x^j \in p} V(x^j) = z(p)^T v w^c w$$

where $v \in S_v$, $w^c \in C_w$, and $w \in O_w$. In RPM, a portfolio p is feasible if $Az(p) \leq B$ holds componentwise for given $A \in \mathbb{R}^{q \times m}$ and $B \in \mathbb{R}^q$ that model some linear constraints. In the stage-gate model, budget and schedule constraints for the development costs and time can be considered in Decision Gates 1–3 to support resource allocation in Development Stages 2–4, respectively.

The overall value of the portfolio depends on the scores and weights employed from the information set. While the resulting value intervals can overlap, some portfolios can be identified as inferior based on dominance. Specifically, portfolio p dominates p' with regard to the information set S iff $V(p, v, w^c, w) \geq V(p', v, w^c, w)$ for all $(v, w^c, w) \in S$ and $V(p, v, w^c, w) > V(p', v, w^c, w)$ for some $(v, w^c, w) \in S$. In other words, portfolio p dominates p' iff

$$\min_{w \in S_w} \left[\underline{V}(p \setminus p', w) - \overline{V}(p' \setminus p, w) \right] \geq 0 \text{ and}$$
$$\max_{w \in S_w} \left[\overline{V}(p \setminus p', w) - \underline{V}(p' \setminus p, w) \right] > 0, \tag{B.4}$$

where

$$\underline{V}(p, w) := \sum_{j=1}^{m} z_j(p) \sum_{o \in O} w_o \underline{v}_o^j \text{ and } \overline{V}(p, w) := \sum_{j=1}^{m} z_j(p) \sum_{o \in O} w_o \overline{v}_o^j.$$

Here, we denote $p \setminus p' := \{x^j \in p | x^j \notin p'\}$. If p is not dominated by any other portfolio, it is a non-dominated portfolio, denoted by p_N. The set of non-dominated portfolios is denoted with P_N. As a rational decision-maker would not choose a portfolio that is dominated, recommendations can be based on the analysis of non-dominated portfolios. A candidate which is included in all (none) of the non-dominated portfolios is called a *core candidate* (an *exterior candidate*). A core candidate should be chosen for the next development stage, whereas an exterior candidate should be terminated.

If there are few feasible portfolios, the non-dominated portfolios can be found by pairwise comparing all feasible portfolios and checking if conditions (B.4) hold. When the number of feasible portfolios increases, exact and approximative algorithms can be applied (see, e.g., Mild et al., 2015).

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